
AI-102 Natural Language Processing (NLP)

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EMERGING &
COLLABORATIVE STUDIES



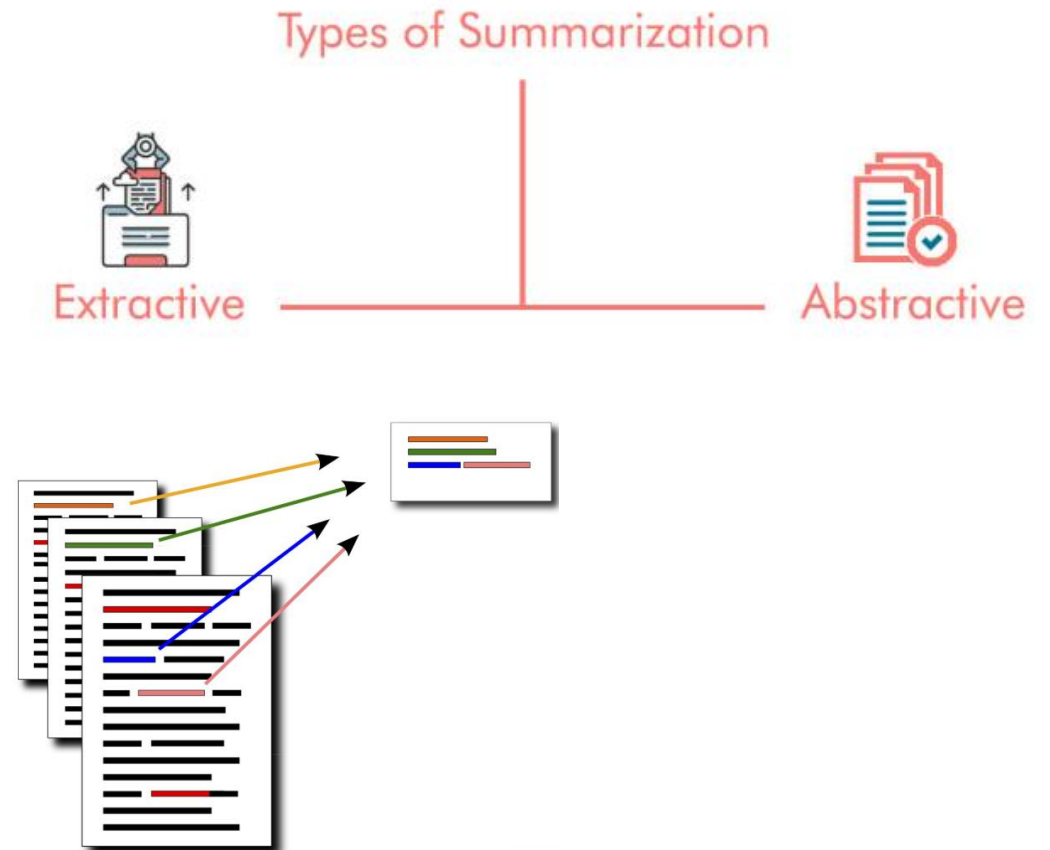


Text Summarization

What's Extractive Summarization

Extractive summarization is a method of automated text summarization that creates a summary by selecting and extracting the most important sentences, phrases, or ideas directly from the original text.

- Builds a summary by choosing the **most important** sentences
- Uses features like:
 - Term frequency
 - Sentence position
 - Similarity to title/topic
- Fast, simple, and widely used (news, document filtering)



Understanding Extractive Summarization

1. Preprocessing

- Tokenization (sentences/words)
- Stopword removal
- Normalization (lowercase)

2. Sentence Scoring

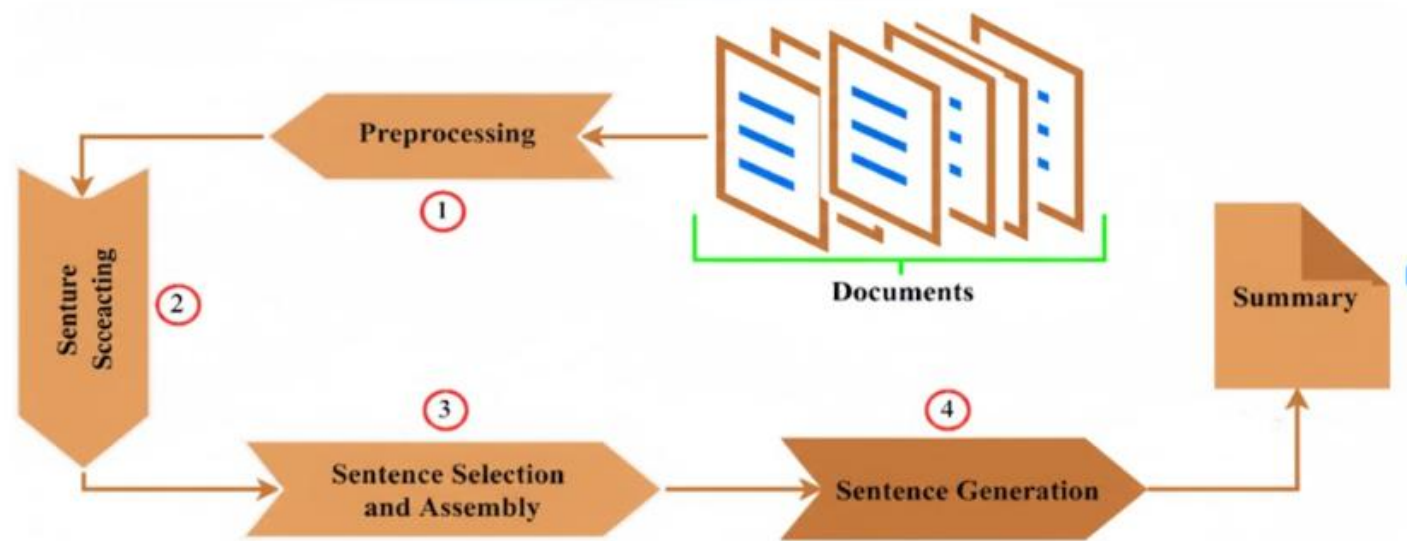
- Count important words
- Consider position
- Measure similarity

3. Sentence Selection

- Pick top-scoring sentences

4. Summary Generation

- Arrange selected sentences coherently



Advanced Extractive Summarization Techniques

1. TextRank (Graph-Based)

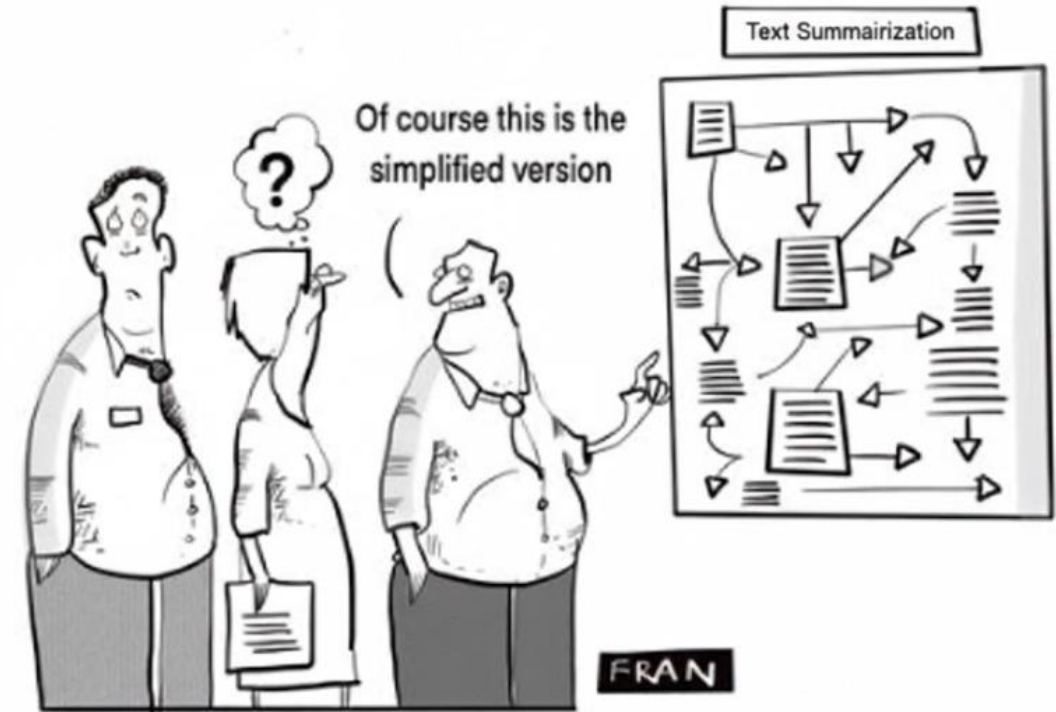
- Build a **similarity graph** of sentences
- Apply PageRank to score sentences
- Select highest-ranked ones

2. Latent Semantic Analysis (LSA)

- Build term-document matrix
- Apply **SVD** to capture hidden semantic structure
- Select sentences linked to main latent topics

3. Supervised Learning

- Train ML model on **labeled summaries**
- Features: term frequency, position, length, semantic similarity
- Classifies sentences as *important* (1) or *not important* (0)



Advantages and Limitation Extractive Summarization



- Simple & easy to implement
- **Fast** → works well for large datasets
- **Accurate to source** (no rewriting)

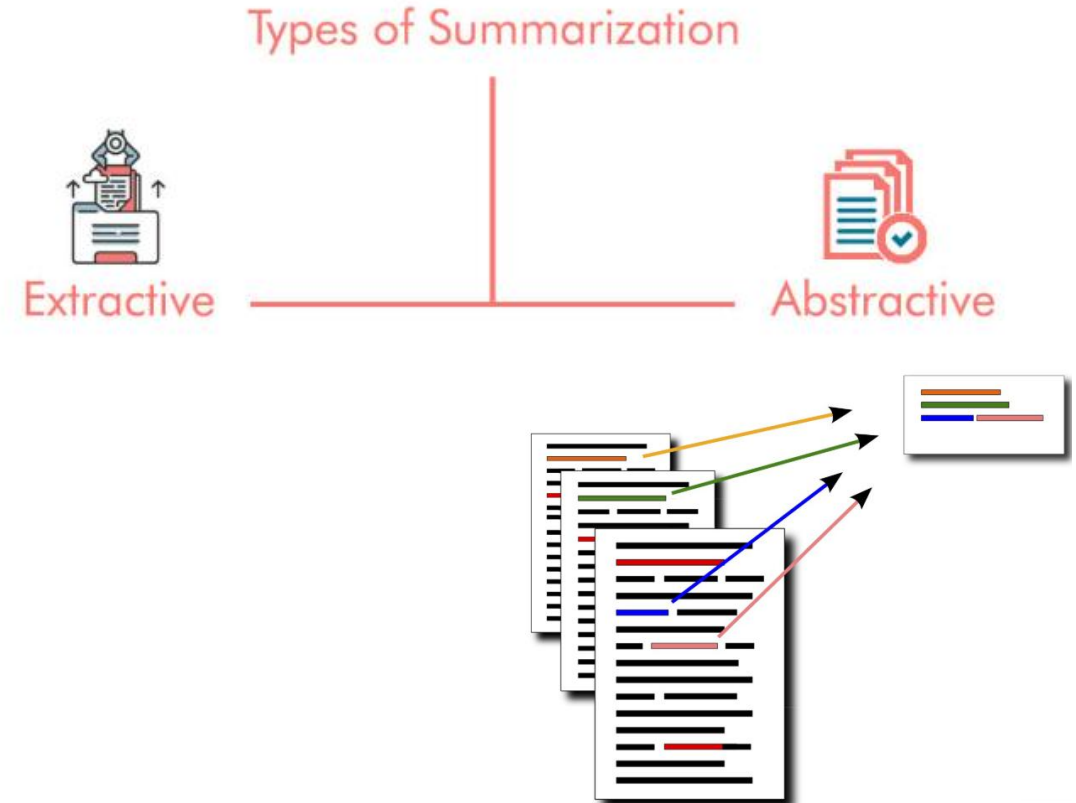


- **Coherence issues:** sentences may not flow well
- **Redundancy:** may pick similar sentences
- **No abstraction:** cannot paraphrase or generate new wording
- Relies on presence of “good” sentences in original text

What's Abstractive Summarization

Abstractive summarization is a technique for summarizing text by using natural language generation to create a new, condensed version of the original content, rather than just pulling sentences directly from the source

- Goes **beyond** picking existing sentences (unlike extractive summarization).
- Tries to summarize text the way **humans** do: shorter, more **readable**, more **informative**
- Especially useful for **long** or **complex** documents.



Understanding Abstractive Summarization

1. Encoder

- Reads the input text.
- Produces a context vector (compressed representation of meaning).

2. Decoder

- Uses the context vector to generate the summary.
- Must be clear, concise, and faithful to the source.

Common architectures:

- RNNs, LSTMs, Transformer-based models
- Trained on large datasets of (document, summary) pairs.



*It's my selfie stick. But I look a lot more
ABSTRACTIVE from a distance*

Advanced Abstractive Summarization Techniques

Transformer-based models

- **BERT** – strong bidirectional encoder; understands context deeply.
- **GPT** – autoregressive generator; good at fluent text generation.
- **BART** – combines bidirectional encoder + autoregressive decoder; fine-tuned for summarization.

Advanced Abstractive Summarization Techniques

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Pointer-Generator Networks

- Mix **extractive** + **abstractive** ideas.
- Can **copy** words directly from the source **or generate** new words.
- Helps keep **technical terms** accurate while improving readability.

Advanced Abstractive Summarization Techniques

Transformer-based models

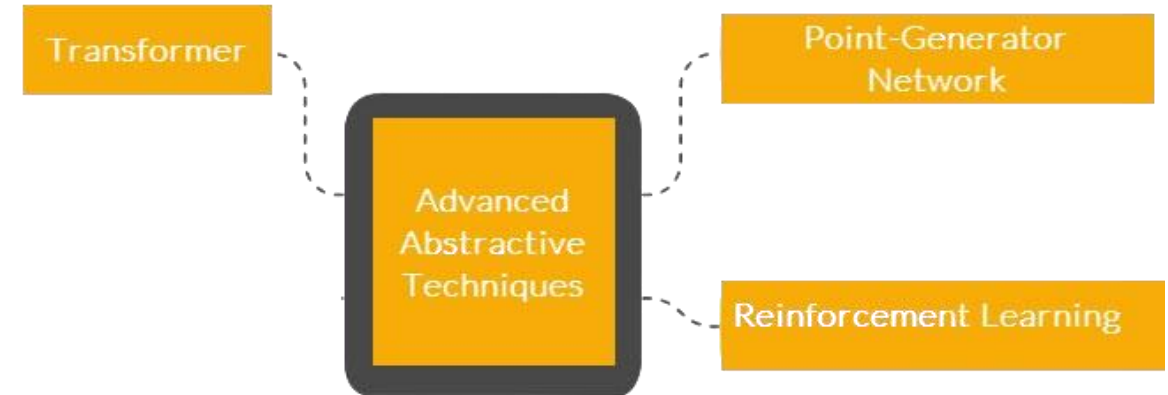
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Reinforcement Learning

- Model receives **rewards** for better summaries (e.g., coherence, relevance).
- Learns to **optimize** its decisions over time based on feedback.







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